

SCIENCE AND TECHNOLOGY OF MATERIALS – TECHNOLOGY OF METALLIC MATERIALS



Fe – Fe₃C Diagram Recap

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- **Rule 3:** If we know T and C_0 , then can determine:
- the weight fraction of each phase.

- Examples:

Consider $C_0 = 35 \text{ wt\% Ni}$

At T_A : Only Liquid (L) present

$$W_L = 1.00, W_\alpha = 0$$

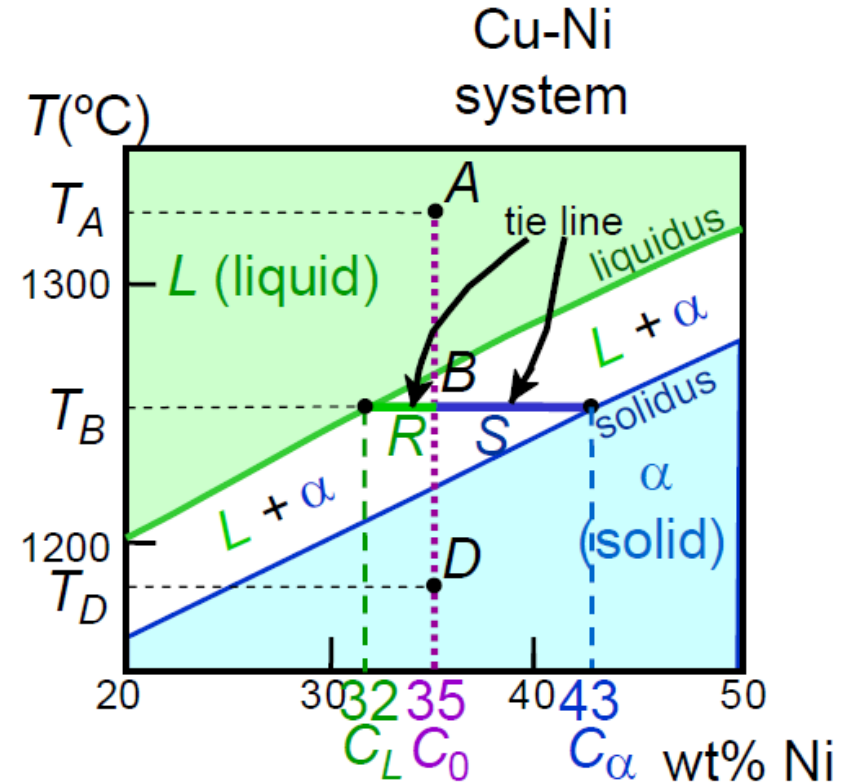
At T_D : Only Solid (α) present

$$W_L = 0, W_\alpha = 1.00$$

At T_B : Both α and L present

$$W_L = \frac{S}{R+S} = \frac{43-35}{43-32} = 0.73$$

$$W_\alpha = \frac{R}{R+S} = 0.27$$



SOLUBILITY DIAGRAMS LEVER RULES



➤ Sum of weight fractions: $W_L + W_\alpha = 1$

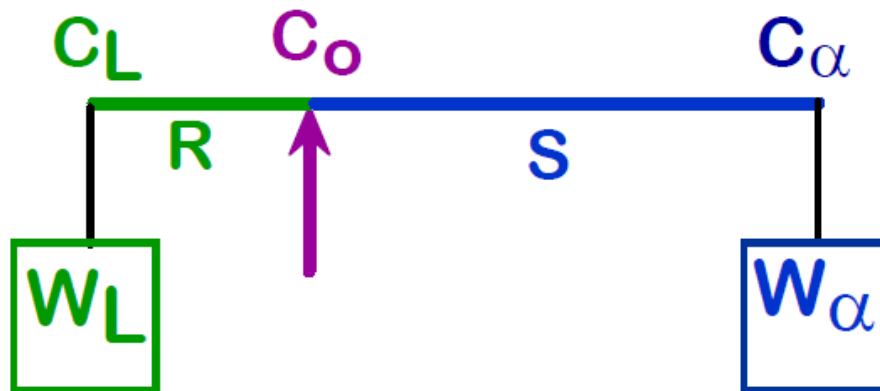
➤ Conservation of mass: $C_O = W_L C_L + W_\alpha C_\alpha$

➤ Combine above equations:

$$W_L = \frac{C_\alpha - C_O}{C_\alpha - C_L} = \frac{\text{S}}{\text{R} + \text{S}}$$

$$W_\alpha = \frac{C_O - C_L}{C_\alpha - C_L} = \frac{\text{R}}{\text{R} + \text{S}}$$

• A geometric interpretation:



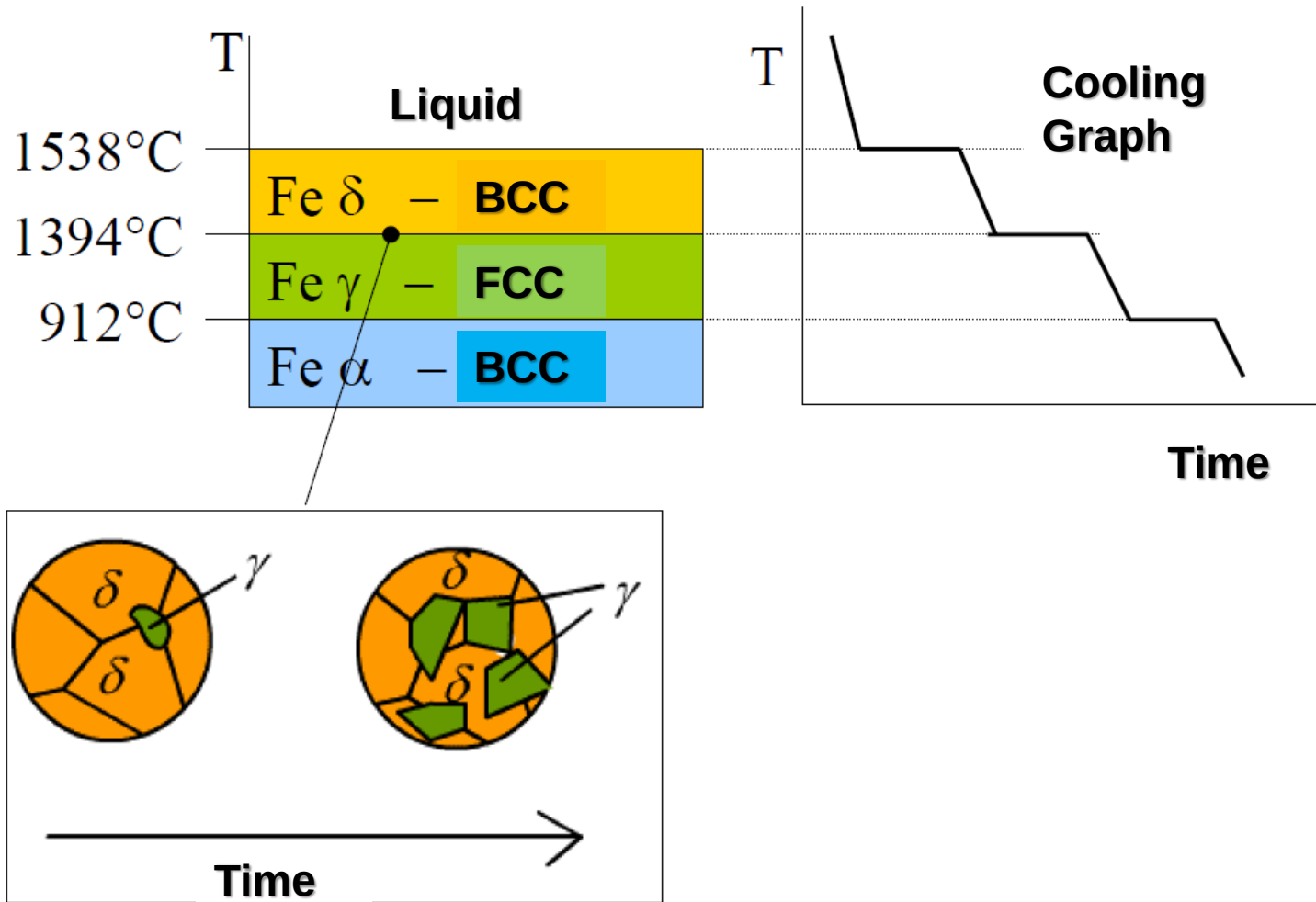
moment equilibrium:

$$W_L R = W_\alpha S$$

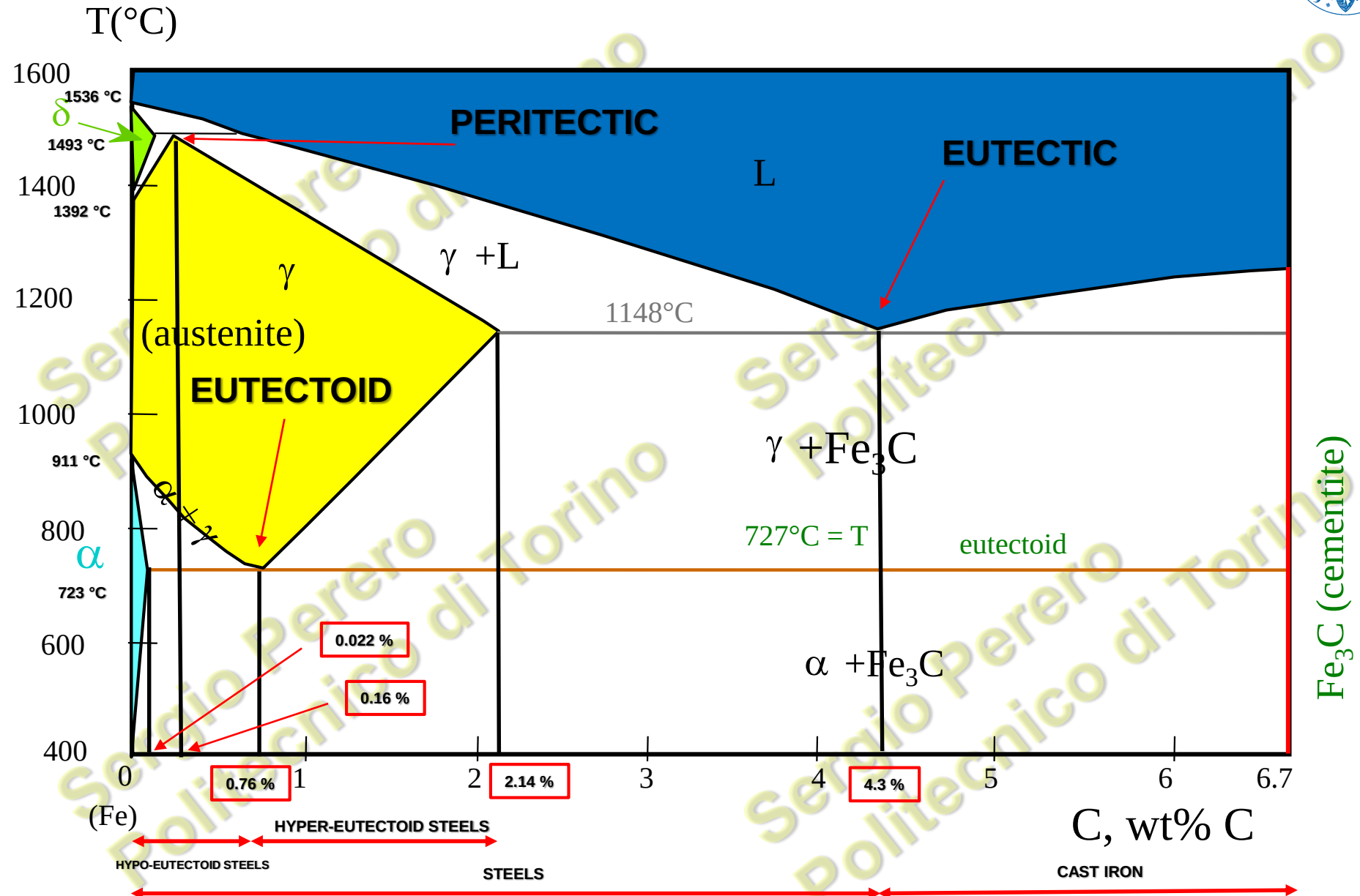
1 - W_α

solving gives Lever Rule

IRON ALLOTROPY

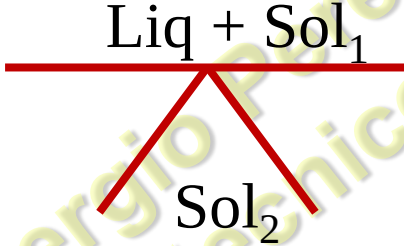
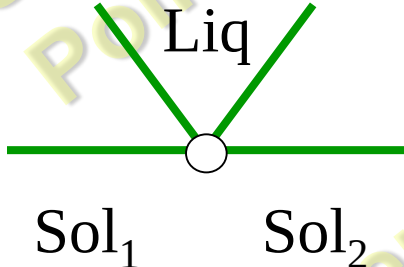
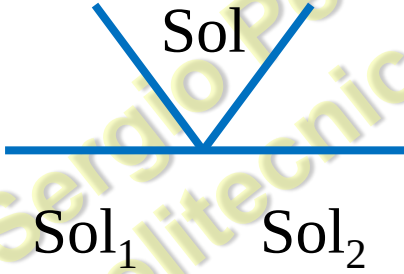


Fe – C AND Fe – Fe₃C PHASE DIAGRAMS



Fe₃C (cementite)

INVARIANT POINTS IN THE FE-C SYSTEM

Invariant point	Reaction	phase diagram	location
PERITECTIC	$\text{Liq} + \text{Sol}_1 \xrightleftharpoons[\text{heating}]{\text{cooling}} \text{Sol}_2$		$C = 0.16 \%C$ $T_p = 1493 ^\circ C$
EUTECTIC	$\text{Liq} \xrightleftharpoons[\text{heating}]{\text{cooling}} \text{Sol}_1 + \text{Sol}_2$		$C = 4.3\%$ $T_{EU} = 1148 ^\circ C$
EUTECTOID	$\text{Sol} \xrightleftharpoons[\text{heating}]{\text{cooling}} \text{Sol}_1 + \text{Sol}_2$		$C = 0.76\%$ $T_{ED} = 723 ^\circ C$

PHASES



➤ **α -ferrite** - solid solution of C in BCC Fe

- Stable form of iron at room temperature.
- Transforms to FCC γ -austenite at 912 °C

➤ **γ -austenite** - solid solution of C in FCC Fe

- Transforms to BCC δ -ferrite at 1395 °C
- Is not stable below the eutectic temperature (727 °C) unless cooled rapidly.

➤ **δ -ferrite** solid solution of C in BCC Fe

- It is stable only at T, >1394 °C. It melts at 1538 °C

➤ **Fe₃C (iron carbide or cementite)**

- This intermetallic compound is metastable at room T. It decomposes (**very slowly, within several years**) into α -Fe and C (graphite) at 650 - 700 °C

➤ **Fe-C liquid solution**

Troostite

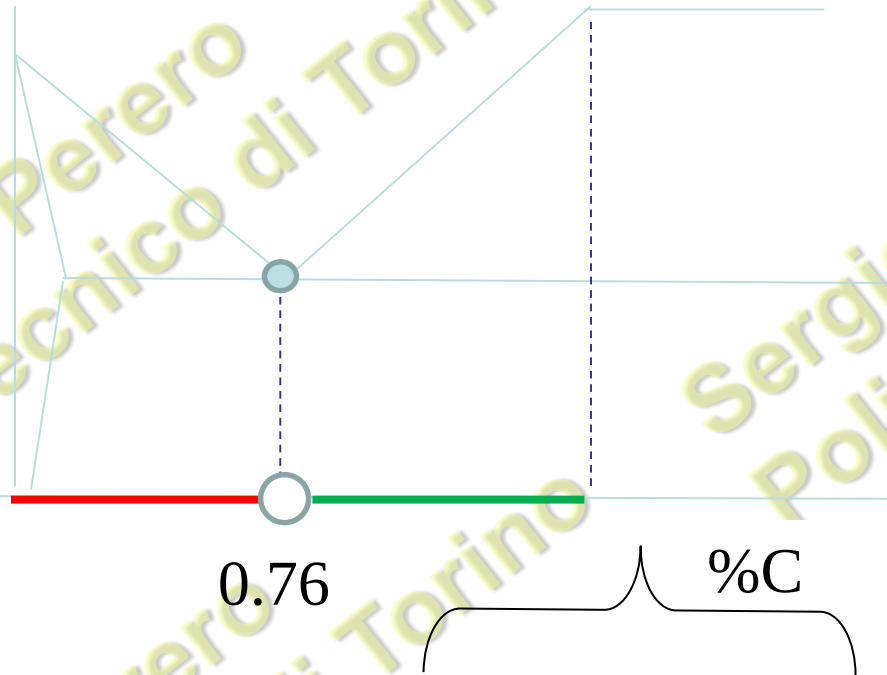
Bainite (upper and lower)

Martensite

Sorbite

Can be obtained by rapid thermal treatments.

CLASSIFICATION OF STEELS AS A FUNCTION OF %C



MICROSTRUCTURE

(PHASES ARE α ; Fe_3C)

Hypo-eutectoid steels:

$\%C < 0.76 \Rightarrow$

ferrite + pearlite

Eutectoid Steels:

$\%C = 0.76 \Rightarrow$

pearlite

Hyper-eutectoid: Steels

$\%C > 0.76 \Rightarrow$

pearlite + cementite

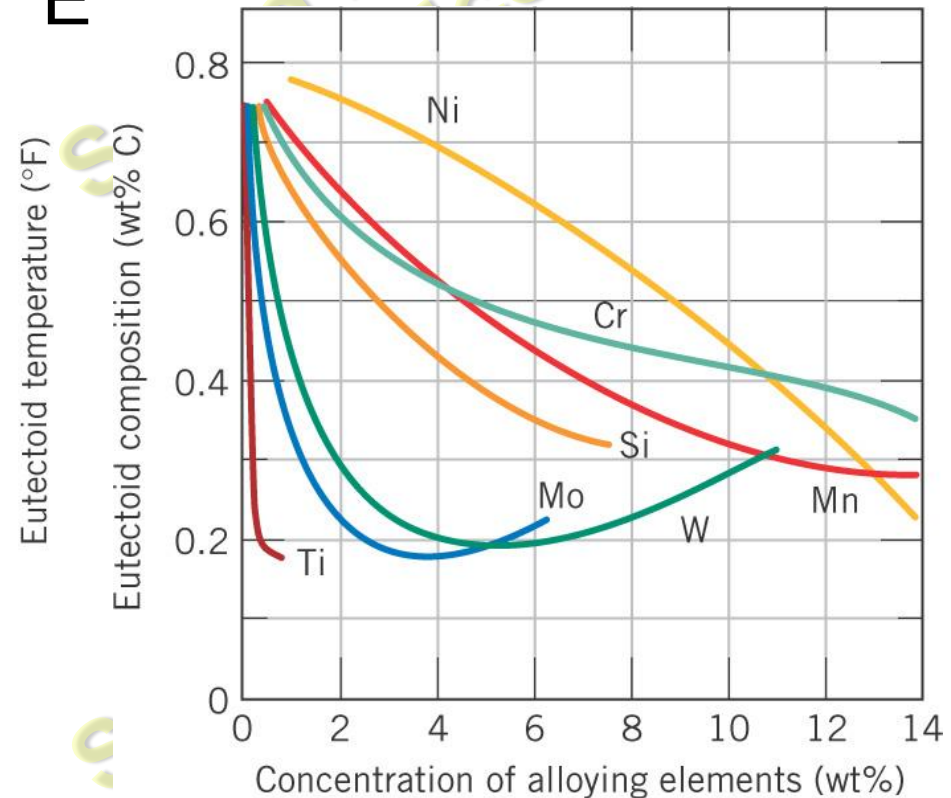
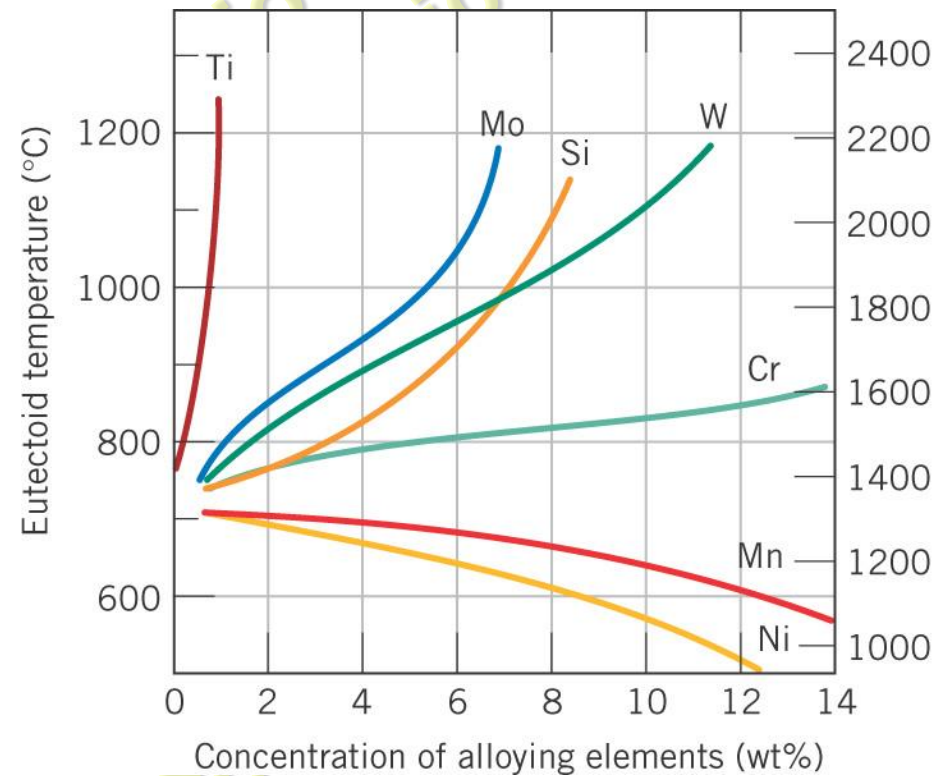
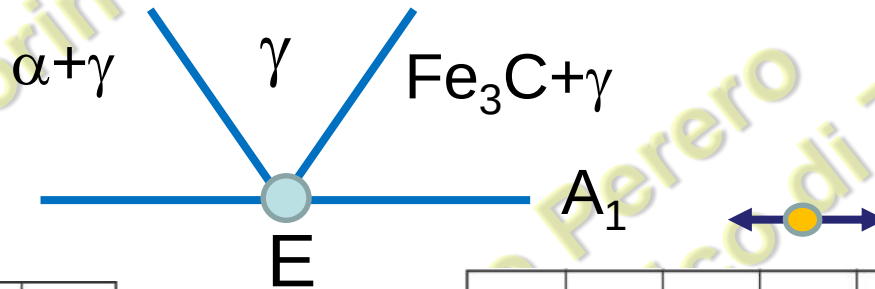
Strong AUSTENITIZING: Ni, Mn, Co (@ T_{ambient})
weak AUSTENITIZING: C, N, Cu

expand the austenite phase field

Strong FERRITIZING: Cr, Si, Al, Mo, Ti, V,

expand the ferrite phase field

INFLUENCE OF ALLOYING ELEMENTS ON A_1 and C_E



QUESTIONS?

THE END

